Practical 5:

Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Code:-

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Student Registration Form</title>

  <style>

    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
      padding: 30px;
    }

    .container {
      width: 500px;
```

```
    max-width: 100%;
    margin: auto;
    background-color: white;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
}

h1 {
    text-align: center;
}

label {
    display: block;
    margin-top: 15px;
    font-weight: bold;
}

input,
select {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #aaa;
    border-radius: 5px;
}

.gender {
```

```
    display: flex;
    gap: 15px;
    margin-top: 8px;
}

.gender input {
    width: auto;
}

.terms {
    margin-top: 15px;
}

.terms input {
    width: auto;
}

button {
    width: 100%;
    margin-top: 20px;
    padding: 12px;
    border: none;
    border-radius: 5px;
    background-color: #007bff;
    color: white;
    font-size: 16px;
    cursor: pointer;
}
```

```
        button:hover {
            background-color: #0056b3;
        }

        .error {
            color: red;
            font-size: 13px;
        }

        #success {
            color: green;
            text-align: center;
            font-weight: bold;
            margin-top: 15px;
        }

    </style>

</head>

<body>

    <div class="container">

        <h1>Student Registration</h1>

        <form id="registrationForm">
```

```
        <!-- Name -->
        <label for="name">Name</label>

        <input type="text" id="name" placeholder="Enter your name">

        <div class="error" id="nameError"></div>


        <!-- Email -->
        <label for="email">Email</label>

        <input type="email" id="email" placeholder="Enter your email">

        <div class="error" id="emailError"></div>


        <!-- Mobile -->
        <label for="mobile">Mobile Number</label>

        <input type="tel" id="mobile" placeholder="Enter 10 digit mobile
number">

        <div class="error" id="mobileError"></div>


        <!-- Password -->
        <label for="password">Password</label>
```

```
        <input type="password" id="password" placeholder="Enter password">

        <div class="error" id="passwordError"></div>


        <!-- Confirm Password -->
        <label for="confirmPassword">Confirm Password</label>

        <input type="password" id="confirmPassword" placeholder="Confirm
password">

        <div class="error" id="confirmPasswordError"></div>


        <!-- Course -->
        <label for="course">Course</label>

        <select id="course">

          <option value="">Select Course</option>

          <option value="B.Tech">B.Tech</option>

          <option value="BCA">BCA</option>

          <option value="MCA">MCA</option>
```

```
    <option value="BBA">BBA</option>

</select>

<div class="error" id="courseError"></div>


<!-- Year -->
<label for="year">Year</label>

<select id="year">

    <option value="">Select Year</option>

    <option value="1">1st Year</option>

    <option value="2">2nd Year</option>

    <option value="3">3rd Year</option>

    <option value="4">4th Year</option>

</select>

<div class="error" id="yearError"></div>


<!-- Gender -->
```

```
<label>Gender</label>

<div class="gender">

  <label>
    <input type="radio" name="gender" value="Male">
    Male
  </label>

  <label>
    <input type="radio" name="gender" value="Female">
    Female
  </label>

  <label>
    <input type="radio" name="gender" value="Other">
    Other
  </label>

</div>

<div class="error" id="genderError"></div>


<!-- Terms -->
<div class="terms">

  <label>
```

```
            <input type="checkbox" id="terms">

            I accept the Terms and Conditions

          </label>

        </div>

        <div class="error" id="termsError"></div>


        <button type="submit">
          Register
        </button>

        <div id="success"></div>

      </form>

    </div>


    <script>


document.getElementById("registrationForm").addEventListener("submit",
function(event) {
```

```
        event.preventDefault();

        // Get values

        let name = document.getElementById("name").value.trim();

        let email = document.getElementById("email").value.trim();

        let mobile = document.getElementById("mobile").value.trim();

        let password = document.getElementById("password").value;

        let confirmPassword =
document.getElementById("confirmPassword").value;

        let course = document.getElementById("course").value;

        let year = document.getElementById("year").value;

        let gender = document.querySelector('input[name="gender"]:checked');

        let terms = document.getElementById("terms").checked;



        // Regular Expressions

        let namePattern = /^[A-Za-z ]{3,30}$/;
```

```
        let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

        let mobilePattern = /^[6-9][0-9]{9}$/;

        let passwordPattern = /^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-
9])(?=.*[@#$%^&*!]).{8,}$/;



        // Clear previous errors

        document.querySelectorAll(".error").forEach(function(error) {
           error.innerHTML = "";
        });

        document.getElementById("success").innerHTML = "";



        let valid = true;



        // Name validation

        if (!namePattern.test(name)) {

           document.getElementById("nameError").innerHTML =
              "Name must contain only letters and spaces (3-30 characters).";
```

```
    valid = false;

}


// Email validation

if (!emailPattern.test(email)) {

    document.getElementById("emailError").innerHTML =
        "Enter a valid email address.";

    valid = false;

}


// Mobile validation

if (!mobilePattern.test(mobile)) {

    document.getElementById("mobileError").innerHTML =
        "Enter a valid 10 digit Indian mobile number.";

    valid = false;

}
```

```
        // Password validation

        if (!passwordPattern.test(password)) {

            document.getElementById("passwordError").innerHTML =
                "Password must be 8+ characters with uppercase, lowercase,
number and special character.";

            valid = false;

        }



        // Confirm password

        if (password !== confirmPassword) {

            document.getElementById("confirmPasswordError").innerHTML =
                "Passwords do not match.";

            valid = false;

        }



        // Course validation
```

```
if (course === "") {

   document.getElementById("courseError").innerHTML =
      "Please select a course.";

   valid = false;

}


// Year validation

if (year === "") {

   document.getElementById("yearError").innerHTML =
      "Please select your year.";

   valid = false;

}


// Gender validation

if (!gender) {

   document.getElementById("genderError").innerHTML =
      "Please select your gender.";
```

```
    valid = false;

}


// Terms validation

if (!terms) {

    document.getElementById("termsError").innerHTML =
        "You must accept the terms and conditions.";

    valid = false;

}


// Final result

if (valid) {

    document.getElementById("success").innerHTML =
        "Registration successful!";

    document.getElementById("registrationForm").reset();

}
```

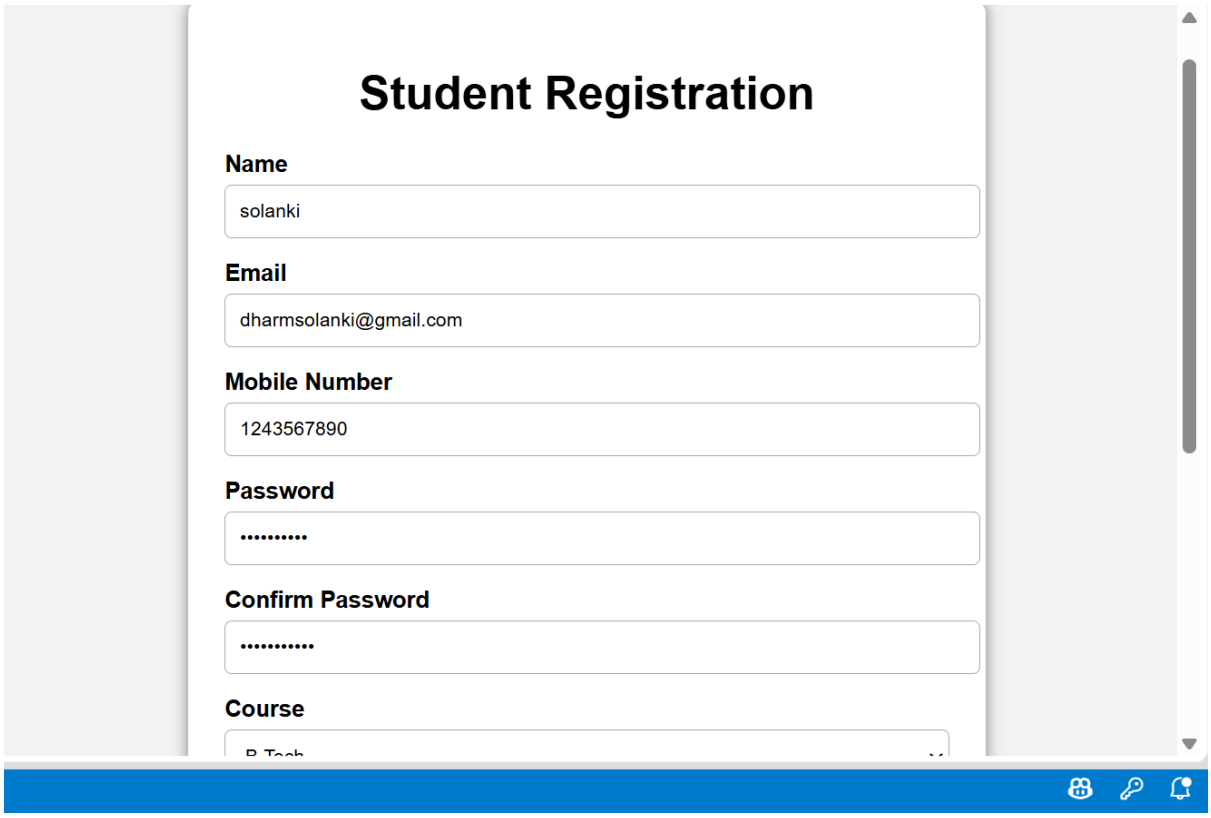
```
    });

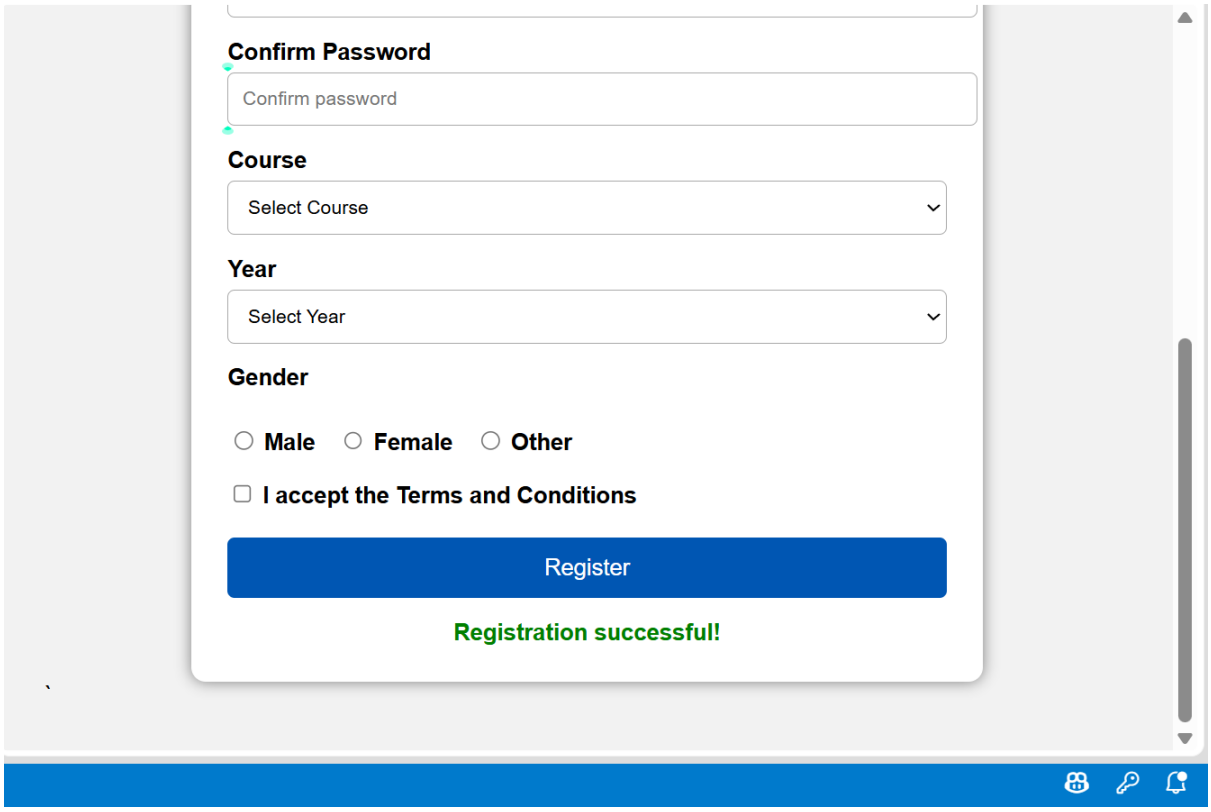
  </script>

</body>

</html>
```

Output:-





Key Questions / Analysis / Interpretation Answers

1. Are correct input types and attributes used?
Yes. The form uses appropriate HTML5 input types such as text, email, tel, and password. <select>, radio buttons, and a checkbox are also used for course, year, gender, and terms.

2. Are validation errors displayed near the relevant fields?
Yes. Each field has a separate error message area, such as nameError, emailError, mobileError, etc. This makes it easy for the user to identify and correct the error.

3. Is password strength checked?
Yes. A Regular Expression checks that the password has:

- Minimum 8 characters
- One uppercase letter
- One lowercase letter
- One number
- One special character

Example: Dharma@123

4. Is the form accessible using keyboard and screen-reader-friendly labels?
Mostly yes. The form uses <label> elements

connected to inputs using for and id, so screen readers can identify the fields. Standard HTML controls also support keyboard navigation. However, accessibility could be improved further by adding required, aria-described by, and aria-live attributes for validation messages.

Tools/Technology to be Used :

HTML5, JavaScript ES5/ES6, browser dev tools